HAZOP Methodology and Risk Ranking

The selected HAZOP node is the product fractionation column that removes naphtha overhead while recovering jet and diesel-range hydrocarbons in the bottoms. This unit was chosen because it is strongly tied to the process control discussion and because it contains several interacting variables: pressure, reflux, reboiler duty, condenser duty, feed rate, and inventory levels that can quickly affect both safety and product quality. The HAZOP is organized by node: feed system, column body, overhead condenser and reflux drum, reboiler system, and bottoms withdrawal.

Table 1 - HAZOP Analysis of Jet Fuel Fractionator

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Instrument Tag	Recommended Action
T-700 Feed	FV-700	low flow	valve fails close, loss of level in D-700 or D-701, low pressure in drums	process upset, jet off spec	4	FI-700 will indicate low flow, naphtha spec allows some contamination from jet	2	4	FI-700	
T-700 Feed	FV-700	high flow	LV-700 or LV-701 fails open or liquid level lost in D-700 or D701	Hydroprocessing reactor section depressures into T-701, possibly causing loss of containment and fire	1	FI-700 will indicate high flow, column pressure indication	2	1	FI-700	provide PSV on column and overhead receiver, high pressure trip and low drum level trip to feed valves
T-700 Feed	TI-700	high temp	inadequate cooling in E-700 or too much heat transfer in E-603	process upset, naphtha off spec	2	TI-700 will indicate high temp	2	2	TI-700	Provide TI's on overhead vapor, reboiler vapor, reflux liquid, bottoms liquid, feed liquid, maybe feed tray provide low level alarms on D700 & D701
T-700 Feed	TI-700	low temp	too much cooling in E-700 or low outlet temperature from E-603	process upset, Jet off spec	3	TI-700 will indicate low temp, column temperatures will warn of upset conditions	2	3	TI-700	^
T-700 Feed	FI-700	loss of containment	flange leak, valve stem leak	venting hydrocarbon, possible fire or explosion	1	equipent design & maintenance	3	2	-	Install automatic ESD isolation on fractionator feed, improve valve/flange mechanical integrity with leak detection, LEL monitor

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Instrument Tag	Recommended Action
T-700 Feed	outside of section scope	variation in composition	temperature excursion in R-601, catalyst deactivation, or temperature/pressure variation in D-700 or D-701	process upset, products off spec	3	temperature, pressure and flow measurements in R-601, regular sampling analysis	4	4	-	
T-700 Overhead	PV-700	low pressure	valve fails open, ambient conditions such as rainfall cause too much cooling in E-701	process upset, naphtha off spec	4	PI-700 indicates pressure	3	4	PI-700	provide offgas flowmeter for material balance and troubleshooting
T-700 Overhead	PV-700	high pressure	valve fails close, ambient conditions cause inadequate cooling in E-701	process upset, jet off spec, pressure build up into loss of containment	1	PI-700 indicates pressure	3	2	PI-700	installation of PRV's and PSV's on column and accumulator, alarm
T-700 Overhead	TI-701	high temp	Ambient conditions cause inadequate cooling in E-701	process upset, naphtha off spec	4	TI-701 indicates temperature	4	4	TI-701	
T-700 Overhead	TI-701	low temp	Ambient conditions cause too much cooling in E-701	process upset, jet off spec	3	TI-701 indicates temperature	1	2	TI-701	provide TI-701 high alarm, consider louvers on E-701
T-700 Overhead	PI-700	loss of containment	flange leak, valve stem leak, external fire	venting non-condensables, possible environmental issue, fire, severe equipment damage	1	unit design & maintenance	4	3	-	provide local LEL alarms where leaks are possible
D-700 Overhead Receiver	D-700	low level	feed loss, cooling loss, high pressure, reboiler heat loss	process upset, jet & naphtha off spec	2	level, flow and pressure indicators	2	2	-	provide level alarms
D-700 Overhead Receiver	D-700	high level	drop in column pressure, TV-700 fails close, LI-702 error, P-700 trips	liquid into fuel gas system, possible loss of containment and fire	3	pressure, temperature, and level indication	4	4	-	provide level alarms
T-700 Liquid Product	TV-700	high flow	TV-700 fails open, column temperature reading incorrect	process upset, naphtha off spec	4	FI-701 will indicate high flow, maintenance	2	4	FI-701	
T-700 Liquid Product	TV-700	low flow	TV-700 fails close, P-700 trips	process upset, jet off spec	3	FI-701 will indicate low flow, maintenance	2	3	FI-701	
T-700 Liquid Product	TI-704	high temp	inadequate cooling in E-701	process upset, naphtha off spec	3	TI-702 indicates temperature, column	2	3	TI-704	

Equipment	Tag	Deviation	Cause	Consequence	Severity	Safeguard Description	Likelihood	Risk	Instrument Tag	Recommended Action
						temperature indication				
T-700 Liquid Product	TI-704	low temp	too much cooling in E-701	process upset, jet off spec	3	TI-702 indicates temperature, column temperature indication	2	3	TI-704	
T-700 Liquid Product	FI-701	loss of containment	flange leak, valve stem leak	venting hydrocarbon, possible fire or explosion	1	unit design & maintenance	5	NH	-	provide LEL monitor near areas of concern
T-700 Liquid Product	outside of section scope	variation in composition	temperature variation in column, reflux valve failure	process upset, product off spec	2	column temperature indication, valve maintenance, product sampling and analysis	2	2	-	temperature & pressure alarms, flow indicators
T-700 Bottom Product	FV-705	high flow	FV-705 fails open, high sump level	possible lost sump level, process upset, Jet off spec	2	FI-705 indicates flow, LI-705 level, maintenance	3	3	FI-702, LI-705	installation of PRV's and PSV's downstream of control valve, alarms
T-700 Bottom Product	FV-705	low flow	FV-705 fails close, low sump level, P-704 trips	possible overfill sump covering reboiler vapor return causing tray damaage, process upset, jet off spec	2	FI-705 indicates flow, LI-705 level, maintenance	3	3	FI-702, LI-705	low level alarms
T-700 Bottom Product	TI-706	high temp	steam header upset or FV-704 malfunction - too much heating in reboiler	process upset, naphtha off spec	3	TI-704 indicates temperature, column temperature indication	2	3	TI-706	
T-700 Bottom Product	TI-706	low temp	steam header upset or FV-704 malfunction - inadequate heating in reboiler	process upset, jet off spec	2	TI-704 indicates temperature, column temperature indication	2	2	TI-706	
T-700 Bottom Product	FI-705	loss of containment	flange leak, valve stem leak, pump seal leak	venting hydrocarbon, possible fire or explosion	1	unit design & maintenance	5	NH	-	provide LEL monitor in area